

Experiment no: 16

Date:

AIM: Network simulation using NS2Objective: To simulate and analyze packet drops, congestion effects, TCP/UDP behaviour and compare data rate and throughput using NS2 in Linux.

a) NS2 simulator Introduction: NS2 is a discrete event simulator for networking research. It supports simulation of wired and wireless network, and protocols like TCP, UDP, AODV etc. simulations are written in OTcl and results can be traced or visualized using NAM.

b) Simulate to find Number of packets Dropped.

script: packet_drop.tcl.

```

set ns [new simulation]
set no [$ns node]
set n1 [$ns node]
$ns duplex-link $no $n1 10Mb 10ms DropTail
set tcp [new Agent [TCP]]; $ns attach-agent $no $tcp
set sink [new Agent [TCPSink]]; $ns attach-agent $n1 $sink.
$ns connect $tcp $sink
set cbr [new Application [Traffic [CBR]]
$cbr attach-agent $tcp; $cbr set packet_size_512; $cbr set interval_0.05
$cbr start
$ns trace-all [open out.tr]
$ns run

```

Command to find dropped packets:

get "drop" out.tr | wc -l

c) Simulate to Find Packets Dropped Due to Congestion

same script as (b) — Use high traffic rate or smaller queue buffer command:

```
get "drop" out.tr | grep "queue" | wc -l
```

d) Simulate to Find packets Dropped by TCP/UDP:

Add UDP Flow:

```
set udp [new Agent|udp]; $ns attach-agent $no $udp
set null [new Agent|null]; $ns attach-agent $n1 $null
$ns connect $udp $null
```

```
set cbr2 [new Applications Traffic|CBR]
```

```
$cbr2 attach-agent $udp; $cbr2 set packetSize 512; $cbr2 set interval 0.05
```

```
$cbr2 start
```

Commands:

```
grep "drop" out.tr | grep "TCP" | wc -l
```

```
grep "drop" out.tr | grep "UDP" | wc -l
```

e) Compare Data Rate and Throughput

To measure:

- Data Rate = total data sent / simulation time
- Throughput = total data received / simulation time.

example command:

```
grep "tr" out.tr | grep "tcp" | wc -l
```

Multiply packet count by packet size to get total bits

\$ ns packets_simulation.tcl

NS2 Simulation – Introduction

Packets Dropped: 25

Packets Dropped by TCP: 14

Packets Dropped by UDP: 11

Packets Dropped due to congestion: 9

Throughput = 0.85 Mbps

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